Program: **Diploma in Computer Science and Engineering**	
Course Code: **5401**	Course Title: **Artificial Intelligence**
Semester: **5**	Credits: **4**
Course Category: **Program Core Course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

Learners will be

Course Prerequisites:

Topic	Course name	Semester
Basic knowledge in mathematics	Mathematics I	1
	Mathematics II	2
Basic concepts of programming	Introduction to IT systems lab	1
	Problem Solving and Programming	2

Course Outcomes

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Artificial Intelligence and the significance of Intelligent Agents in AI.	15	Understanding
CO2	Apply Mathematical logic for knowledge representation of AI	15	Applying
CO3	Practice various search techniques in AI	16	Applying
CO4	Describe learning and expert systems.	14	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the fundamental Concepts of Artificial Intelligence**		
M1.01	Explain about AI, foundations of AI and its history	2	Understanding
M1.02	Discuss applications of AI.	3	Understanding
M1.03	Explain Turing Test.	1	Understanding
M1.04	Explain AI problems	3	Understanding
M1.04	Describe agents and environments.	2	Understanding
M1.05	Identify structure of agents	2	Understanding
M1.06	Recognize types of agents.	2	Understanding
Contents: **Artificial Intelligence concepts-** Foundations of AI, History of AI **Applications of AI** - Autonomous planning and scheduling- Game playing- Autonomous control- Diagnosis- Logistics Planning- Robotics- Language understanding and problem solving. **Intelligent Agents**- Agents and Environments, Structure of Agent - Types of agents. **Examples of AI Problems**: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem			
CO2	**Apply Mathematical logic for knowledge representation of AI**		
M2.01	Summarize Ontologies, foundations of knowledge representation and reasoning	3	Understanding

M2.02	Describe the representation and reasoning about objects, relations, events, actions, time and space	3	Understanding
M2.03	Demonstrate Propositional Logic	2	Applying
M2.04	Demonstrate First Order Logic	2	Applying
M2.05	Explain Soundness and Completeness,	2	Understanding
M2.06	Explain Forward and Backward chaining	3	Understanding
	Series Test – I	1	
Contents: **Knowledge representation and reasoning**-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. **Logic and Inferences:** Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining.			
CO3	**Practice various search techniques in AI**		
M3.01	Summarise uninformed and informed search	1	Understanding
M3.02	Implement Breadth First Search and Depth First Search	3	Applying
M3.03	Implement Best first search, Hill Climbing, A* & Beam Search	3	Applying
M3.04	Solve optimal paths - Branch & Bound, Divide & Conquer approaches, Beam Stack Search	3	Applying
M3.05	Describe the Problem Decomposition techniques - Goal Trees, AO*, Rule Based Systems	3	Understanding
M3.06	Compare and summarize game playing algorithms - Minimax, AlphaBeta	3	Applying

Contents:

Uninformed Search: Breadth First Search, Depth First Search

Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm

Beam Search

Finding optimal paths -Branch & bound, Divide & Conquer approaches

Problem Decomposition: Goal Trees, AO*, Rule Based System.

Game Playing: Minimax Algorithm, AlphaBeta Algorithm.

CO4	**Describe Learning and Expert systems**		
M4.01	Describe Domains, Forward and Backward Search	3	Understanding
M4.02	Explain Goal Stack Planning, Plan Space Planning,	2	Understanding
M4.03	Explain Graphplan, Constraint Propagation	3	Understanding
M4.04	Explain expert system	2	Understanding
M4.05	Describe Knowledge Representation in expert systems	2	Understanding
M4.06	Describe Expert system tools – MYCIN – EMYCIN.	2	Understanding
	Series Test – II	1	

Contents:

Planning and Constraint Satisfaction: Domains, Forward and Backward Search, Goal Stack Planning, Plan Space Planning, GraphPlan, Constraint Propagation**.**

Expert Systems: Definition – Features – Architecture – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN.

Text / Reference:

T/R	Book Title / Author
T1	Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill
R2	Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India)
R3	S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, SecondEdition, Pearson Education 2
R4	Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann
R5	Donald A.Waterman, ‘A Guide to Expert Systems’, Pearson Education

Online Resources:

Sl. No.	Website Link
1	https://www.tutorialspoint.com/virtualization2.0/index.htm
2	https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
3	https://onlinecourses.swayam2.ac.in/arp19_ap79/preview